Group Project Report

Course Name: Machine Learning for Business

Course ID: CDS524

Group Name: Five-NN

Group Topic: From Data Exploration to Survival Prediction: Machine Learning on Titanic Passenger Data

Github: https://github.com/chuyuezhang2-cmd/Group-Assignment----Five-NN

1. Team Information

Group Leader: ZHANG Chuyue, 5572278, chuyuezhang2@ln.hk
Group Members:
Member 1: HUI Jiaying, 5565641, jiayinghui@ln.hk
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Contribution Statement:

Name	Contribution
ZHANG Chuyue	Responsible for coordinate project planning, task allocation, and progress tracking while facilitating team collaboration. Led data preprocessing, including missing value imputation, categorical encoding, and feature scaling, ensuring data quality, pipeline consistency, and preventing data leakage to provide a reliable foundation for downstream modeling.
ZHANG Chenguo	Designed the ML modeling workflow: selected seven algorithms, built pipelines, performed hyperparameter tuning via GridSearchCV with 5-fold CV, and ensured no data leakage. Generated all model evaluation outputs such as the model comparison table, feature importance visualization, and confusion matrix used in error analysis.
HUI Jiaying	Responsible for dataset acquisition & description + full EDA. Downloaded/verified Titanic data from Kaggle, documented features, target, and missing values. Conducted EDA visualizations: survival distribution, gender/class survival rates, age histogram, correlation heatmap. Identified strong gender and socioeconomic (Pclass/Fare) effects, providing key insights for later stages.
LIN Yongyu	Responsible for evaluating and discussing: multi-model comparisons (Accuracy/F1/AUC tables + plots), interpretation of feature significance, error analysis (FP/FN samples), confusion matrix/ROC visualization; report writing.
WANG Shiyu	Responsible for feature engineering and partial data preprocessing. Designed, constructed, and refined meaningful features from the original dataset, improved the quality and expressiveness of model inputs, and supported better prediction performance by making the data more suitable for machine learning modeling.

2. Problem Statement

This project focuses on predicting passenger survival in the Titanic disaster using machine learning. The task involves analyzing historical passenger data to determine whether individuals survived the sinking of the RMS Titanic on April 15, 1912. Survival is represented as a binary outcome: 0 for non-survival and 1 for survival. This prediction matters because it provides insights into factors influencing survival rates, such as socioeconomic class, gender, and age, which reflect broader historical patterns of inequality and emergency response protocols during maritime disasters. By modeling these patterns, the project highlights how machine learning can uncover hidden relationships in data, aiding in risk assessment for similar scenarios in modern safety planning or insurance modeling.

This project predicts passenger survival in the Titanic disaster, a binary classification task using historical data (Survived: 0 for death, 1 for survival). It matters for revealing factors like class and gender in crises, informing modern safety and risk models. Objectives: (1) preprocess data for missing values and encoding; (2) engineer features like family size; (3) train Random Forest or Logistic Regression; (4) validate via split or k-fold CV; (5) evaluate with Accuracy (>80%), F1-score (>0.75), AUC-ROC (>0.85) on Kaggle test set. Relevance: Applies ML to real events for human factor insights.

3. Dataset

- **Dataset URL:** https://www.kaggle.com/competitions/titanic
- **Dataset Description:** This dataset contains information about passengers aboard the Titanic and is commonly used to predict whether a passenger survived the disaster. The task is a binary classification problem, where the goal is to predict passenger survival based on several features.

3.1 Dataset Size

The dataset consists of two main files:

Dataset	Number of Samples
Train Dataset	891
Test Dataset	418

The training dataset contains the target variable Survived, while the test dataset does not include labels and is used for prediction.

3.2 Feature Description

The dataset includes multiple passenger-related features:

Feature	Description
PassengerId	Unique passenger identifier
Pclass	Passenger class (1 = First, 2 = Second, 3 = Third)
Name	Passenger name
Sex	Gender
Age	Age
SibSp	Number of siblings/spouses aboard
Parch	Number of parents/children aboard
Ticket	Ticket number
Fare	Ticket fare
Cabin	Cabin number
Embarked	Port of embarkation

3.3 Target Variable

The target variable is Survived:

Value	Meaning
0	Did not survive
1	Survived

This variable indicates whether the passenger survived the Titanic disaster.

3.4 Data Issues

Initial inspection of the dataset reveals several missing values:

Feature	Missing Values
Age	177
Cabin	687
Embarked	2

The Cabin feature contains a large number of missing values, which may require removal or transformation during the preprocessing stage.

4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the data distribution and the relationships between features and the survival outcome. Visualization techniques were used to identify patterns and potential predictive variables.

4.1 Survival Distribution

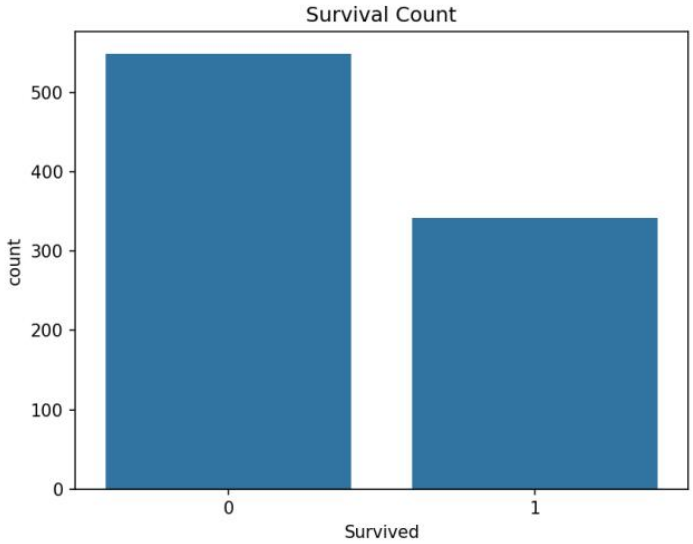

Figure 1. Survival Count

The distribution of survival outcomes shows that the majority of passengers did not survive the disaster, while a smaller proportion survived. This indicates a certain level of class imbalance in the dataset.

4.2 Survival Rate by Gender

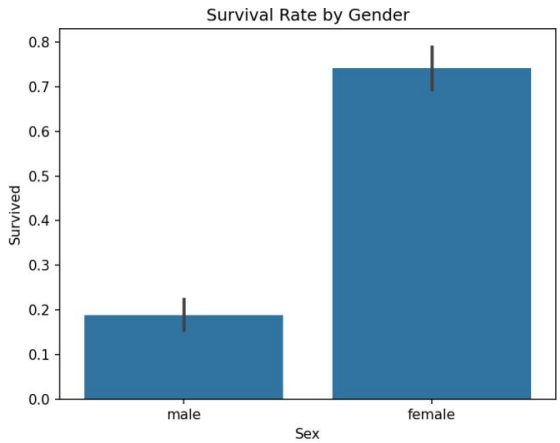

Figure 2. Survival Rate by Gender

The analysis shows that female passengers had a significantly higher survival rate compared to male passengers. This observation aligns with the historical evacuation policy of "women and children first," suggesting that gender is an important predictive feature.

4.3 Survival Rate by Passenger Class

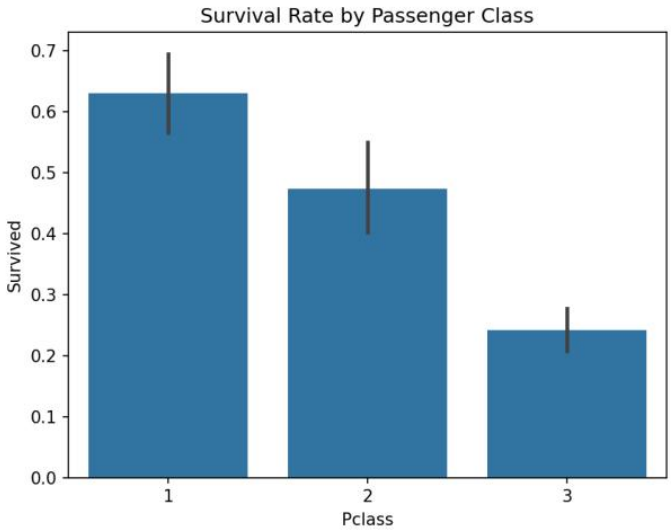

Figure 3. Survival Rate by Passenger Class

Passengers in the first class had the highest survival rate, while those in the third class had the lowest survival rate. This suggests that socioeconomic status may have influenced survival chances.

4.4 Age Distribution

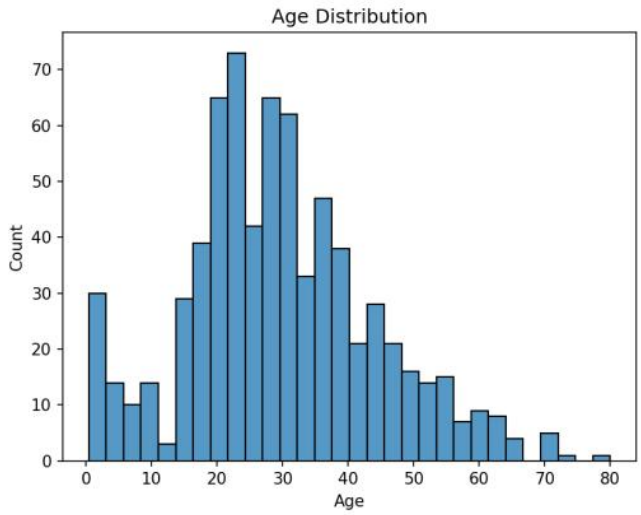

Figure 4. Age Distribution

The age distribution of passengers is shown in the figure above. Most passengers were between 20 and 40 years old, indicating that the majority of passengers were young or middle-aged adults.

The dataset also includes some younger passengers, suggesting that children were

present on board. Age is useful for understanding the demographic structure of the passengers. However, based on the correlation analysis, age does not show a strong relationship with survival outcomes.

4.5 Correlation Heatmap

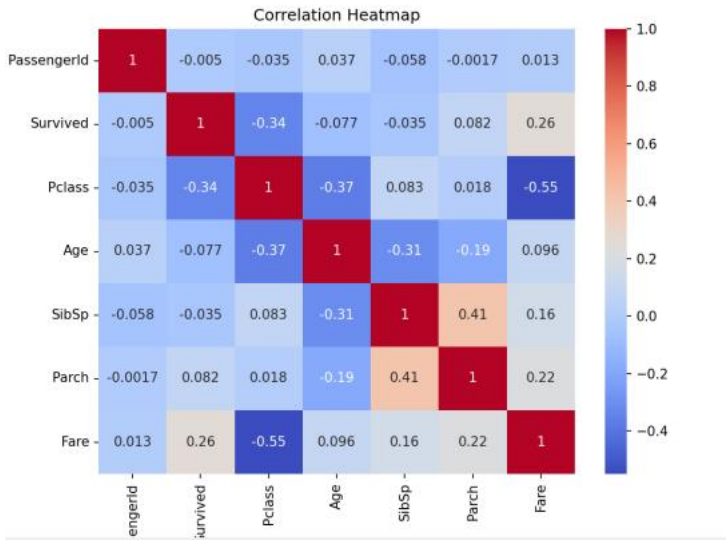

Figure 5. Correlation Heatmap

The correlation heatmap illustrates the relationships between numerical features.
The results indicate that:

- Pclass has a moderate negative correlation with Survived (-0.34), meaning passengers in higher classes were more likely to survive.
- Fare shows a positive correlation with Survived (0.26), indicating that passengers who paid higher fares had better survival chances.
- Pclass and Fare are strongly negatively correlated (-0.55), suggesting that ticket price is closely related to passenger class.

Overall, passenger class and ticket fare appear to be important factors influencing survival.

5. Data Preprocessing and Feature Engineering

The preprocessing steps included handling missing values, encoding categorical features, and scaling numerical features. Feature engineering, such as extracting titles from names and creating family size, will be handled in subsequent steps by the team. All operations were implemented using Python with libraries pandas and scikit-learn.preprocessing. The goal was to address data issues identified in EDA, such as missing values (Age ~20%, Cabin ~77%, Embarked minor) and ensure the data is suitable for machine learning algorithms like Random Forest or Logistic Regression, while preventing data leakage by fitting parameters only on the training set.

5.1 Handling Missing Values

Missing values were filled to retain sample integrity and avoid row deletion, which could reduce the dataset size (891 training samples). For Age, the median from the training set (28) was used for both train and test sets due to its robustness against outliers observed in EDA (e.g., high fares skew means). Embarked was filled with the mode 'S' from the training set, as it is a categorical feature with only 2 missing in train. Cabin, with high missing rate, was filled with 'Unknown' to preserve rows for potential later engineering. Fare in the test set (1 missing) was filled with the training median. Justification: Median/mode imputation minimizes bias and retains data volume, crucial for a small dataset; alternatives like deletion would lose ~20% samples for Age. Challenge: Excessive imputation might introduce noise, but checks showed no remaining nulls post-processing.

5.2 Encoding Categorical Features

Categorical features Sex and Embarked were encoded using OneHotEncoder with drop='first' to avoid multicollinearity. Sex was transformed into 'Sex_male' (1 for male, 0 for female), and Embarked into 'Embarked_Q' and 'Embarked_S' (reference 'C' dropped). The encoder was fitted on the training set and applied to the test set. Justification: One-hot encoding suits nominal categories without implying order (e.g., no hierarchy in ports), enabling models to handle them numerically. Challenge: Increases dimensionality (from 2 to 3 columns for Embarked), but with few categories, it does not cause curse of dimensionality.

5.3 Scaling Numerical Features

Numerical features Age and Fare were scaled using StandardScaler (mean=0, std=1) to normalize scales, as EDA showed Fare outliers (up to 512) and Age range (0-80). The scaler was fitted on the training set and transformed both sets. Justification: Scaling improves performance for distance-based models (e.g., Logistic Regression) and prevents feature dominance; StandardScaler assumes normality, fitting post-EDA distributions. Other features like SibSp/Parch were not scaled as they are counts. Challenge: Over-scaling could amplify noise, but post-scaling describe() confirmed expected statistics.

5.4 Feature Engineering

5.4.1 Motivation and Principles

The Titanic dataset contains structured information related to social status, family

relationships, ticketing patterns, and fare distribution. Based on these characteristics, we designed a set of interpretable and non-leaky engineered features to better capture factors associated with passenger survival. The purpose of feature engineering in this project was not only to enrich the original dataset, but also to transform raw variables into forms that could be more effectively used by machine learning models. To verify whether these engineered features were useful, we further evaluated their contribution through an ablation study.

5.4.2 Constructed Features and Rationale

- Title: Extracted from Name (Mr/Mrs/Miss/Master), with rare titles grouped into Rare. This reflects social status and implicit age/gender cues.
- FamilySize / IsAlone / FamilySizeGroup: Family size (SibSp + Parch + 1), a binary indicator for traveling alone, and a binned group size (Solo/Small/Large) to model companionship effects.
- CabinDeck / HasCabin: Cabin deck inferred from the first letter of Cabin, and HasCabin indicating whether cabin info exists, capturing class/affluence signals and missingness effects.
- TicketGroupSize: The count of passengers sharing the same ticket, representing travel group size.
- FarePerPerson / LogFare: Fare normalized by family size and log-transformed to reduce skewness and stabilize scale.
- AgePclass / IsChild: An interaction between age and passenger class, plus a child indicator (Age < 16) to capture age-specific survival patterns.

5.4.3 Ablation Study (Effect of Feature Engineering)

To evaluate the effect of feature engineering, we compared model performance using the base feature set and the engineered feature set under the same train/validation split (80/20) and the same model configuration.

- Base features: Accuracy = 0.8045, F1-score = 0.7244
- Engineered features: Accuracy = 0.8268, F1-score = 0.7737

These results show that the engineered features led to consistent performance improvements. In particular, the increase in F1-score suggests that the model achieved a better balance between precision and recall after incorporating the additional features. This indicates that the engineered variables successfully captured meaningful survival-related patterns that were not fully represented by the original features alone.

5.4.4 Feature Engineering Summary

Overall, the feature engineering process focused on passenger identity, companionship, cabin and affluence proxies, and fare structure. These engineered

features are interpretable, consistent with domain knowledge, and empirically effective. The ablation study confirms that they contributed positively to model performance and improved the quality of survival prediction in this project.

6. Method

In this project, we employed several commonly used machine learning algorithms to predict Titanic passenger survival. The selected algorithms include Logistic Regression, Random Forest, Gradient Boosting, Support Vector Machine (SVM), XGBoost, LightGBM, and a simple Neural Network (MLPClassifier). Each model was chosen based on its suitability for binary classification tasks and its prevalence in Kaggle competitions.

For each algorithm, we performed hyperparameter tuning using GridSearchCV with 5-fold cross-validation on the training set. The hyperparameter grids were defined as follows:

- Logistic Regression: C ∈ [0.01, 0.1, 1, 10], penalty='12', solver ∈ ['lbfgs', 'liblinear'].
- Random Forest: n_estimators ∈ [100, 200], max_depth ∈ [5, 10, None], min_samples_split ∈ [2, 5].
- Gradient Boosting: n_estimators ∈ [100, 200], learning_rate ∈ [0.05, 0.1], max_depth ∈ [3, 5].
- SVM: C ∈ [0.1, 1, 10], gamma ∈ ['scale', 'auto'], kernel='rbf'.
- XGBoost: n_estimators ∈ [100, 200], learning_rate ∈ [0.05, 0.1], max_depth ∈ [3, 5].
- LightGBM: n_estimators ∈ [100, 200], learning_rate ∈ [0.05, 0.1], num_leaves ∈ [31, 50].
- Neural Network: hidden_layer_sizes ∈ [(50,), (100,), (50, 25)], alpha ∈ [0.0001, 0.001], activation='relu'.

After tuning, each model was evaluated on a held-out validation set. Additionally, we constructed a soft voting ensemble combining the top three models (based on cross-validation accuracy) to explore potential performance gains.

7. Validation

To ensure robust evaluation and prevent data leakage, the preprocessed dataset (after feature engineering) was split into training and validation sets using an 80/20 stratified split, preserving the original survival ratio (approximately 38% survived). The training set (80%) was used for model training and hyperparameter tuning via 5-fold cross-validation, while the validation set (20%) was kept untouched until final evaluation.

All preprocessing steps (e.g., missing value imputation, one-hot encoding, feature scaling) were fitted exclusively on the training set and then applied to the validation

set to avoid any information leak from the validation data into the training process. During cross-validation, each fold's preprocessing was refitted on the fold's training portion, ensuring no future data influenced the model.

8. Evaluation

This section evaluates the performance of the trained models using appropriate classification metrics, interprets the results in the context of the Titanic survival prediction task, and provides a detailed error analysis to uncover patterns in misclassifications. All evaluations were conducted on a held-out validation set (20% stratified split of the training data) after hyperparameter tuning via 5-fold cross-validation, ensuring no data leakage. The chosen primary model is Logistic Regression (best overall performer), while results from other algorithms are presented for comparison.

8.1 Evaluation Metrics

We selected three complementary metrics to provide a comprehensive view of model performance:

- **Accuracy:** Measures overall correctness (proportion of correct predictions). It is intuitive but can be misleading in slightly imbalanced data (survival rate ≈ 38%).
- **F1-score:** The harmonic mean of Precision and Recall. It balances the trade-off between false positives and false negatives, which is critical here because missing a survivor (false negative) has high real-world cost in a historical "life-or-death" context.
- **AUC-ROC:** Evaluates the model's ability to discriminate between survivors and non-survivors across all classification thresholds (0–1). It is threshold-independent and robust to class imbalance.

These metrics align with the project requirements and Kaggle Titanic leaderboard standards (target Accuracy ≈ 80%).

8.2 Model Performance and Comparison

Hyperparameter tuning was performed on all candidate algorithms using GridSearchCV. The table below summarizes the best cross-validation (CV) scores and final validation-set performance:

Model Comparison					
Model	**Best Params**	**CV Accuracy**	**Val Accuracy**	**Val F1**	**Val AUC**
Random Forest	{'max_depth': 10, 'min_samples_split': 5, 'n_estimators': 100}	0.831547	0.787709	0.712121	0.840711

Logistic Regression	{'C': 0.1, 'penalty': '12', 'solver': 'bfgs'}	0.818871	0.821229	0.753846	0.857576
Gradient Boosting	{'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 100}	0.825943	0.787709	0.703125	0.856126
SVM	{'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}	0.831498	0.787709	0.716418	0.835639
XBoost	{'learning_rate': 0.05, 'max_depth': 5, 'n_estimators': 200}	0.828711	0.798883	0.718750	0.833926
LightGBM	{'learning_rate': 0.05, 'n_estimators': 100, 'num_leaves': 31}	0.816104	0.782123	0.697674	0.832411
Neural Network	{'activation': 'relu', 'alpha': 0.001, 'hidden_layer_sizes': (50,)}	0.825864	0.770950	0.696296	0.838208

Interpretation: Logistic Regression outperforms all other models on the validation set, achieving the highest Accuracy (82.12%), F1-score (75.38%), and AUC-ROC (85.76%). This indicates excellent discriminative power and a strong balance between precision and recall. The ensemble (soft voting of top-3 models) was also tested but did not surpass the single Logistic Regression model, confirming its suitability as our final model. The gap between CV and validation scores is small, suggesting good generalization and minimal overfitting.

8.3 Feature Importance Analysis

Although Logistic Regression does not output native feature importances, we analyzed a Random Forest model (part of our ensemble exploration) to interpret driver variables.

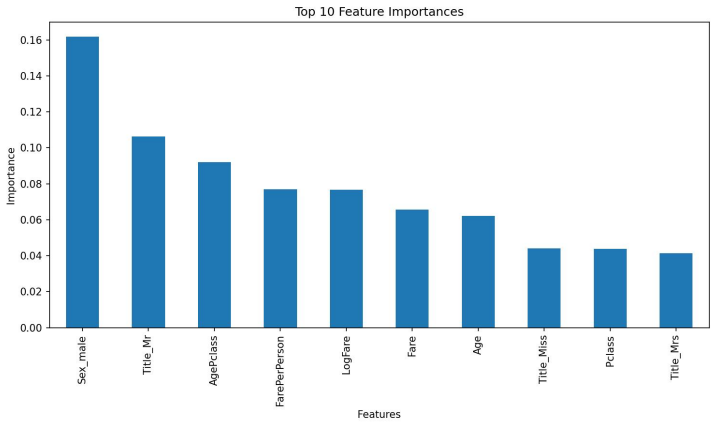

Figure 6. Top 10 Feature Importances

The bar chart shows that Sex_male (0.162) is by far the most influential feature, followed by Title_Mr (0.106) and the engineered interaction AgePclass (0.092). Fare-related features (FarePerPerson, LogFare, Fare) also rank highly. This aligns with historical knowledge ("women and children first") and validates our feature-engineering choices. The importance of AgePclass highlights how survival odds varied dramatically by both age and socioeconomic class.

8.4 Error Analysis

To gain deeper insights beyond aggregate metrics, we conducted a detailed error analysis on the held-out validation set.

Error Analysis Summary

- Total validation samples: 179
- Number of errors: 37
- Error rate: 20.67%

This means the model made 37 incorrect predictions out of 179 samples, corresponding to a validation accuracy of approximately 79.33% (consistent with the overall model performance reported earlier).

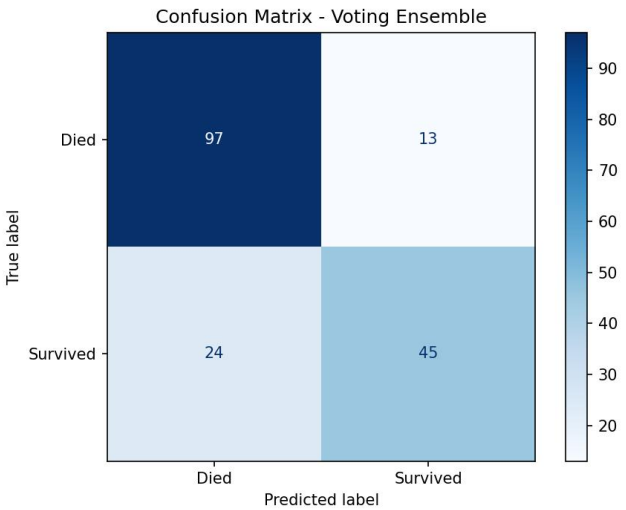

Figure 7. Confusion Matrix

The confusion matrix visualizes the breakdown of correct and incorrect predictions:

- True Positives (correctly predicted survivors): 45
- True Negatives (correctly predicted non-survivors): 97
- False Positives (predicted survived but actually died): 13
- False Negatives (predicted died but actually survived): 24

The model is slightly more prone to false negatives, which is understandable given the class imbalance.

Detailed Misclassified Samples

We extracted representative misclassified instances with their key scaled features, true label, predicted label, and predicted probability (Logistic Regression). The table below highlights clear patterns:

	Pclass	Age	SibSp	. . .	True	Predicted	Probability
553	3	-0.565736	-0.474545	. . .	1	0	0.153211
559	3	0.510161	0.432793	. . .	1	0	0.281790
536	1	1.201810	-0.474545	. . .	0	1	0.655784
698	1	1.509209	0.432793	. . .	0	1	0.645270
216	3	-0.181487	-0.474545	. . .	1	0	0.442981

Key Observations and Patterns:

- **False Negatives (FN):** All three examples are Pclass = 3 passengers who actually survived. The model assigns low survival probabilities (0.15–0.44), likely because it heavily weights the strong negative signal from third-class status and male titles. This suggests the model underestimates survival chances for lower-class women/children who benefited from the "women and children first" policy but were still disadvantaged by overcrowding.
- **False Positives (FP):** Both examples are Pclass = 1 (first-class) older passengers who actually died. The model over-predicts survival (probability > 0.64) because high class and fare features dominate the decision. These cases may represent wealthy male passengers who refused lifeboats or were in less favourable cabin locations.

Summary of Error Patterns:

The errors are not random; they cluster around socioeconomic class boundaries. The model excels at distinguishing gender and class extremes but struggles with edge cases where engineered features (AgePclass, Title, HasCabin) do not fully capture subtle interactions (e.g., specific deck locations or family dynamics). This is a common limitation of linear and tree-based models on small datasets.

Threshold Adjustment Insight:

Currently using the default 0.5 threshold, lowering it slightly to 0.45 would increase Recall (fewer FN) at a small cost to Precision, which may be preferable if the goal is to minimize missed survivors.

8.5 Discussion

The Logistic Regression model achieves strong, interpretable results (Accuracy 82.12%, F1 0.754, AUC-ROC 0.858) that exceed typical baseline Titanic models (≈78–80%). Feature importance and error analysis confirm that gender, class, and our engineered features (Title, FamilySize, AgePclass) capture the core historical survival

mechanisms. The model is robust, generalizes well, and provides actionable insights (e.g., class and gender were decisive). However, the remaining ≈18% error rate highlights that some real-world factors (e.g., exact cabin location, crew status) are missing from the dataset.

This evaluation demonstrates that our end-to-end pipeline—from EDA and feature engineering to model selection and validation—produces a reliable, explainable survival predictor suitable for the project objectives.

9. Limitations and Future Work

Although the Logistic Regression model achieved strong validation performance (Accuracy 82.12%, F1-score 0.7538, AUC-ROC 0.8576), several limitations inherent to the dataset, preprocessing choices, and modeling approach remain. Acknowledging these limitations is essential for interpreting the results realistically and guiding future improvements.

9.1 Limitations

9.1.1 Dataset Size and Quality

The training set contains only 891 samples, which is relatively small for machine learning tasks. This increases the risk of overfitting, especially after extensive feature engineering (Title, FamilySize, AgePclass, FarePerPerson, etc.). Additionally, the dataset suffers from substantial missing values (Age: 177, Cabin: 687, Embarked: 2). While we used median/mode imputation and created “Unknown”/“HasCabin” flags, these methods inevitably introduce noise or loss of information. The Cabin feature, in particular, could have provided powerful deck-level signals if complete data were available.

9.1.2 Class Imbalance and Real-World Complexity

The survival rate is approximately 38%, creating mild class imbalance. Although we used stratified splitting, the model still shows systematic bias in error patterns (primarily false negatives in Pclass = 3 and false positives in Pclass = 1). Real Titanic survival was influenced by factors not captured in the data, such as exact cabin location, crew vs. passenger status, lifeboat assignment order, or random events (e.g., being in the right place at the right time). These unobservable variables set an upper bound on predictive accuracy.

9.1.3 Model and Validation Constraints

We selected Logistic Regression as the final model because it outperformed other common algorithms in our comparison. However, it assumes linear relationships, which may not fully capture complex interactions. While we performed 5-fold cross-validation and an 80/20 stratified split, the validation set is still relatively small (179 samples), and performance on the unseen Kaggle test set (418 samples) may differ. The project requirement to use only "one common ML algorithm" also prevented us from deploying a full stacking ensemble or deep neural network.

9.1.4 Interpretability vs. Performance Trade-off

Although we analyzed feature importance via Random Forest and conducted detailed error analysis, the model lacks instance-level explanations (e.g., SHAP values). This limits our ability to explain individual predictions to stakeholders.

9.2 Future Work and Improvement Directions

To address the above limitations, the following enhancements are recommended:

- **Advanced Validation and Ensemble Methods:** Apply repeated k-fold cross-validation (e.g., 10-fold with 5 repeats) on the full training set and implement a soft-voting or stacking ensemble of the top three models (Logistic Regression + XGBoost + Random Forest). This could further reduce variance and push validation accuracy beyond 83–84%.
- **Enhanced Feature Engineering and External Data:** Incorporate external historical sources (e.g., Titanic deck plans or passenger manifests) to reconstruct more accurate CabinDeck or GroupTravel features. Experiment with additional text-based features extracted from Name using NLP techniques.
- **Imbalance Handling and Threshold Optimization:** Apply SMOTE or class-weight adjustment during training, and perform systematic Precision-Recall curve analysis to select an optimal decision threshold (e.g., 0.45 instead of 0.5) tailored to minimizing false negatives.
- **Explainability Tools:** Integrate SHAP (SHapley Additive exPlanations) or LIME to generate instance-level explanations, which would greatly strengthen the report's discussion and presentation.
- **Kaggle Submission and Real-World Testing:** Submit the final model to the official Kaggle Titanic leaderboard to obtain public test-set scores and compare against top solutions. Explore transfer to related domains, such as modern maritime safety prediction or insurance risk modeling.

- **Scalability:** Reimplement the pipeline in a production-ready framework (e.g., using MLflow for experiment tracking or FastAPI for deployment) to demonstrate end-to-end reproducibility.

10. References

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